

DIGITAL LOGIC DESIGN

(CE_118)

CHAPTER 2:

Storage Components

(part_2)

Storage components

- Storage components store data and perform some simple operations.
- Storage components include:
 - registers
 - counters
 - register files
 - queues
 - stacks
- Combinatorial and storage components are used for construction of:
 - datapaths
 - controllers
- Main subsystems of modern processors and other microchips.

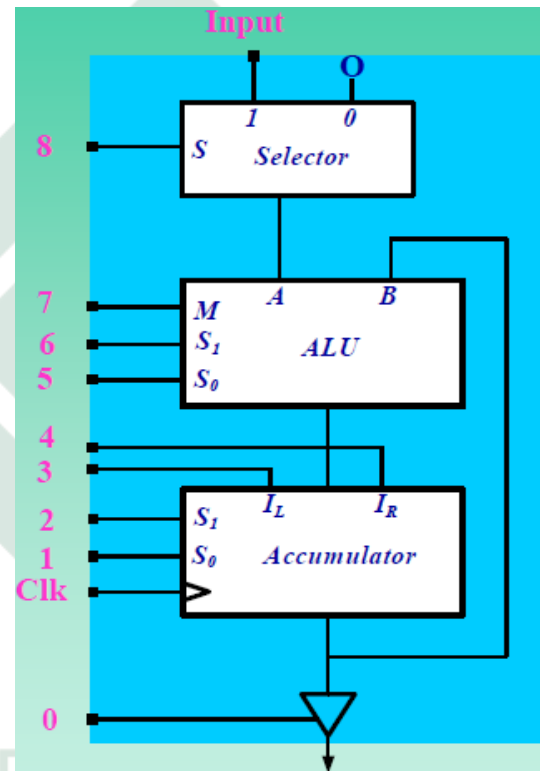
Simple datapath with One accumulator

- Datapath are used for temporary variable storage and operation execution

$$sum = \sum_{i=1}^{100} x_i$$

```

sum = 0
loop:
  for i = 1 to 100
    sum = sum + xi
  end loop
  
```



Datapath schematic

M	S ₁	S ₀	ALU Operations
0	0	0	Complement A
0	0	1	AND
0	1	0	EX-OR
0	1	1	OR
1	0	0	Decrement A
1	0	1	Add
1	1	0	Subtract
1	1	1	Increment A

ALU operations

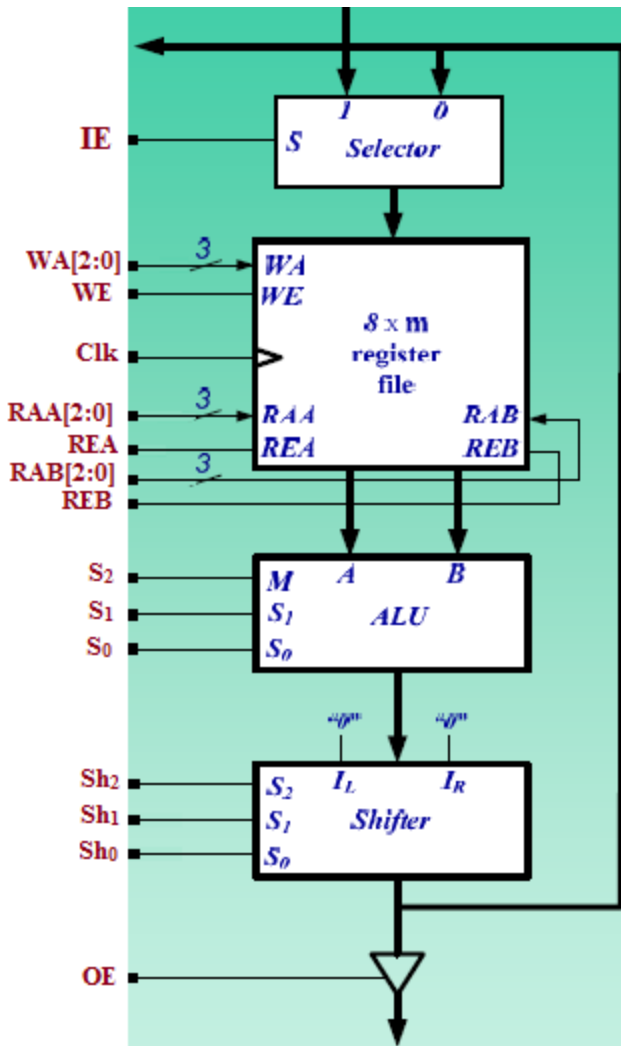
Present state		Operation	Next state			
S ₁	S ₀		Q ₃	Q ₂	Q ₁	Q ₀
0	0	No change	Q ₃	Q ₂	Q ₁	Q ₀
0	1	Load input	I ₃	I ₂	I ₁	I ₀
1	0	Shift left	Q ₂	Q ₁	Q ₀	I _R
1	1	Shift right	I _L	Q ₃	Q ₂	Q ₁

Accumulator (SRwPL) operations

Control word

8	7	6	5	4	3	2	1	0
Input select		ALU controls		Shift values		Accumulator controls		Out enable

Datapath with 3 port register-file



M	S_1	S_0	ALU Operations
0	0	0	Complement A
0	0	1	AND
0	1	0	EX-OR
0	1	1	OR
1	0	0	Decrement A
1	0	1	Add
1	1	0	Subtract
1	1	1	Increment A

ALU operations

S_2	S_1	S_0	Shift Operations
0	0	0	Pass
0	0	1	Pass
0	1	0	Not used
0	1	1	Not used
1	0	0	Shift left
1	0	1	Rotate left
1	1	0	Shift right
1	1	1	Rotate right

Shifter operations

Control word

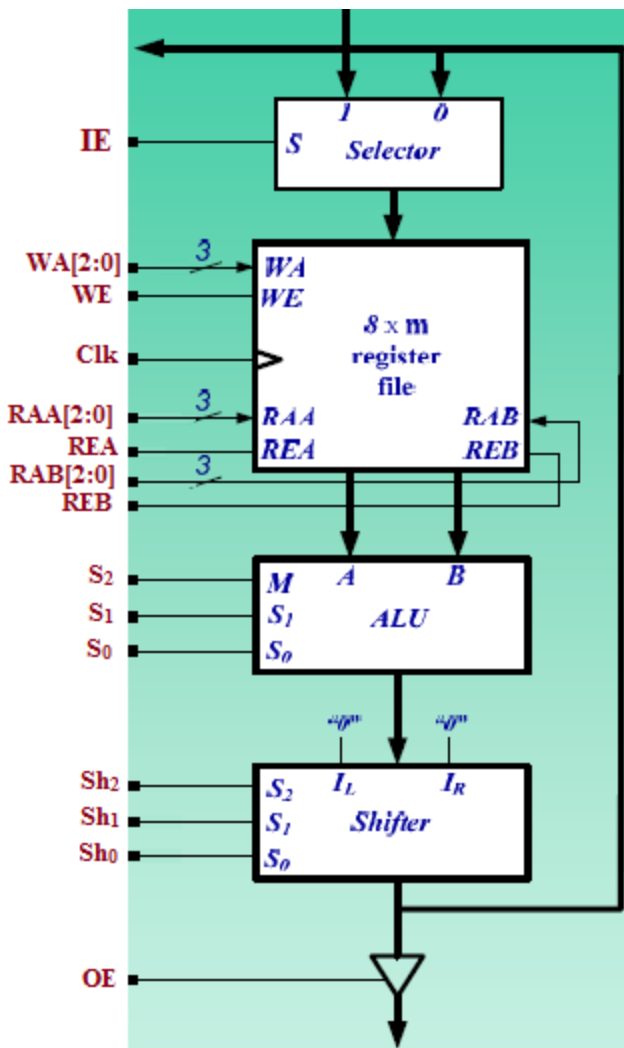
19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IE	Write address			Read address A			Read address B			ALU operation			Shifter operation			OE			

One's-count algorithm

- **Example:** One's-counter implementation.
- **Problem:** Using a datapath with a 3 port register file, design a one's counter that will count the number of ones in an input dataword , and return the result after completion

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Datapath with 3 port register-file



Datapath
schematic

<i>M</i>	<i>S</i> ₁	<i>S</i> ₀	ALU Operations
0	0	0	Complement A
0	0	1	AND
0	1	0	EX-OR
0	1	1	OR
1	0	0	Decrement A
1	0	1	Add
1	1	0	Subtract
1	1	1	Increment A

ALU operations

<i>S</i> ₂	<i>S</i> ₁	<i>S</i> ₀	Shift Operations
0	0	0	Pass
0	0	1	Pass
0	1	0	Not used
0	1	1	Not used
1	0	0	Shift left
1	0	1	Rotate left
1	1	0	Shift right
1	1	1	Rotate right

Shifter operations

Control word

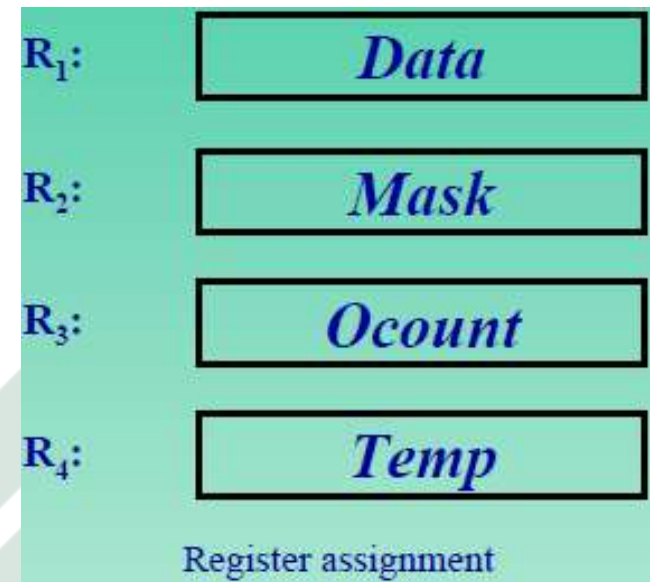
19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IE	Write address				Read address A				Read address B				ALU operation			Shifter operation			OE

One's-count algorithm

```

1. Data := Inport
2. Ocount := 0
3. Mask := 1
while Data ≠ 0 repeat
  4. Temp := Data AND Mask
  5. Ocount := Ocount + Temp
  6. Data := Data >> 1
end while
7. Output := Ocount
Basic algorithm for one's counter

```



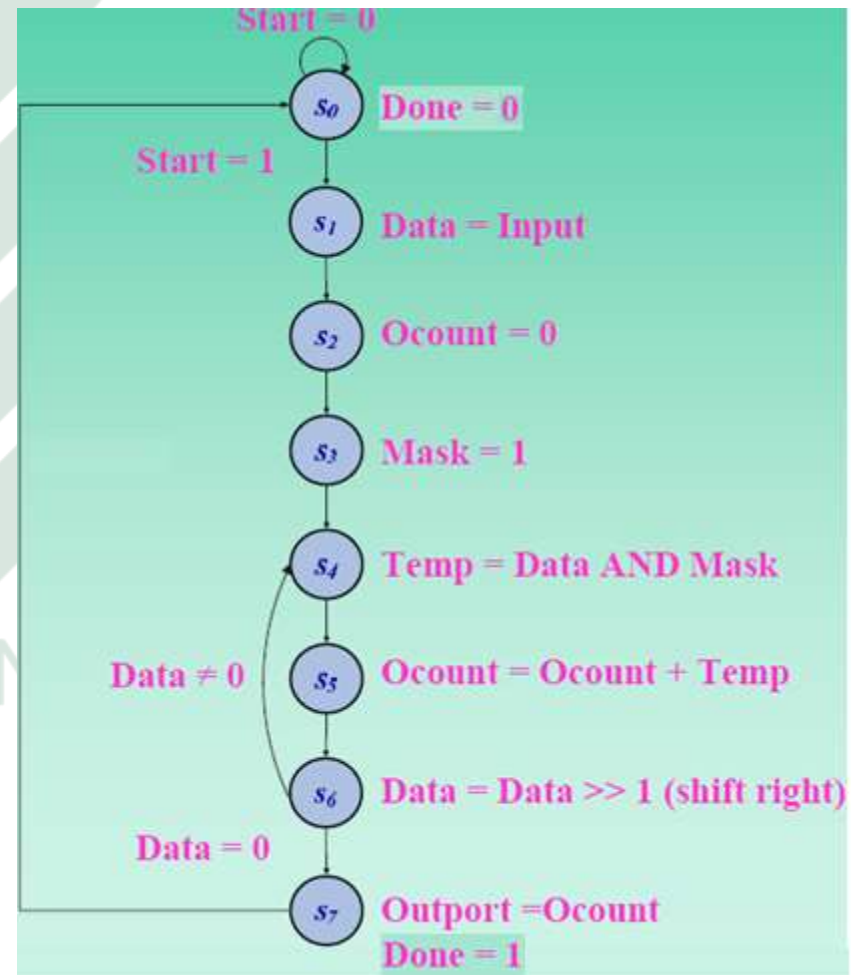
Control words	IE	Write address	Read address A	Read address B	ALU operation	Shifter operation	OE
1	1	R ₁	X	X	X	X	0
2	0	R ₃	0	0	Add	Pass	0
3	0	R ₂	0	X	Increment	Pass	0
4	0	R ₄	R ₁	R ₂	AND	Pass	0
5	0	R ₃	R ₃	R ₄	Add	Pass	0
6	0	R ₁	R ₁	0	Add	Shift right	0
7	0	None	R ₃	0	Add	Pass	1

Repeated
while
Data ≠ 0

Control words for one's counter

FSM representation of One's-counter

- State lasts for a clock cycle.
- In each state the datapath executes the statement indicated on its right side.



Sách sai tín hiệu Done

Next-state logic for One's- counter

States	$Q_2Q_1Q_0$	Start, (Data = 0)			
		00	01	10	11
s_0	0 0 0	000	000	001	001
s_1	0 0 1	010	010	010	010
s_2	0 1 0	011	011	011	011
s_3	0 1 1	100	100	100	100
s_4	1 0 0	101	101	101	101
s_5	1 0 1	110	110	110	110
s_6	1 1 0	100	111	100	111
s_7	1 1 1	000	000	000	000

Next-state table

$$Q_2(next) = Q_2'Q_1Q_0 + Q_2Q_1' + Q_2Q_0'$$

$$Q_1(next) = Q_1'Q_0 + Q_2'Q_1Q_0' + (Data=0)Q_1Q_0'$$

$$Q_0(next) = Q_2'Q_1Q_0' + Q_2Q_1'Q_0' + StartQ_2'Q_0' + (Data=0)Q_2Q_0'$$

Next-state equations

Start, (Data = 0)		$Q_2 = 0$				$Q_2 = 1$			
		Q_1Q_0	00	01	11	10	00	01	11
00	000	010	100	011	101	110	000	100	
01	000	010	100	011	101	110	000	111	
11	001	010	100	011	101	110	000	111	
10	001	010	100	011	101	110	000	100	

$Q_2Q_1Q_0$

Karnaugh map

Output logic for One's- counter controller

				Write address				Read address A				Read address B				ALU operations			Shift operations			
State	Q ₂ Q ₁ Q ₀	IE	WA ₂ WA ₁ WA ₀ WE	RAA ₂ RAA ₁ RAA ₀ REA	RAB ₂ RAB ₁ RAB ₀ REB	S ₂ S ₁ S ₀	Sh ₂ Sh ₁ Sh ₀	OE														
S ₀	0 0 0	0	X X X 0	X X X 0	X X X 0	X X X	X X X	0														
S ₁	0 0 1	1	0 0 1 1	X X X 0	X X X 0	X X X	X X X	0														
S ₂	0 1 0	0	0 1 1 1	0 0 0 1	0 0 0 1	1 0 1	0 0 0	0														
S ₃	0 1 1	0	0 1 0 1	0 0 0 1	X X X 0	1 1 1	0 0 0	0														
S ₄	1 0 0	0	1 0 0 1	0 0 1 1	0 1 0 1	0 0 1	0 0 0	0														
S ₅	1 0 1	0	0 1 1 1	0 1 1 1	1 0 0 1	1 0 1	0 0 0	0														
S ₆	1 1 0	0	0 0 1 1	0 0 1 1	0 0 0 1	1 0 1	1 1 0	0														
S ₇	1 1 1	0	X X X 0	0 1 1 1	0 0 0 1	1 0 1	0 0 0	1														

$$IE = Q'_2 Q'_1 Q_0$$

$$RAA_2 = 0$$

$$RAB_2 = Q'_1 Q_0$$

$$S_2 = Q_1 + Q_0$$

$$WA_2 = Q'_1 Q'_0$$

$$RAA_1 = Q_2 Q_0$$

$$RAB_1 = Q'_1 Q'_0$$

$$S_1 = Q'_2 Q_0$$

$$WA_1 = Q_2 Q_0 + Q_2 Q'_1$$

$$RAA_0 = Q_2$$

$$RAB_0 = 0$$

$$S_0 = 1$$

$$Sh_2 = Sh_1 = Q_2 Q_1 Q'_0$$

$$WA_0 = Q'_1 Q_0 + Q_1 Q'_0$$

$$REA = Q_2 + Q_1$$

$$REB = Q_2 + Q_1 Q'_0$$

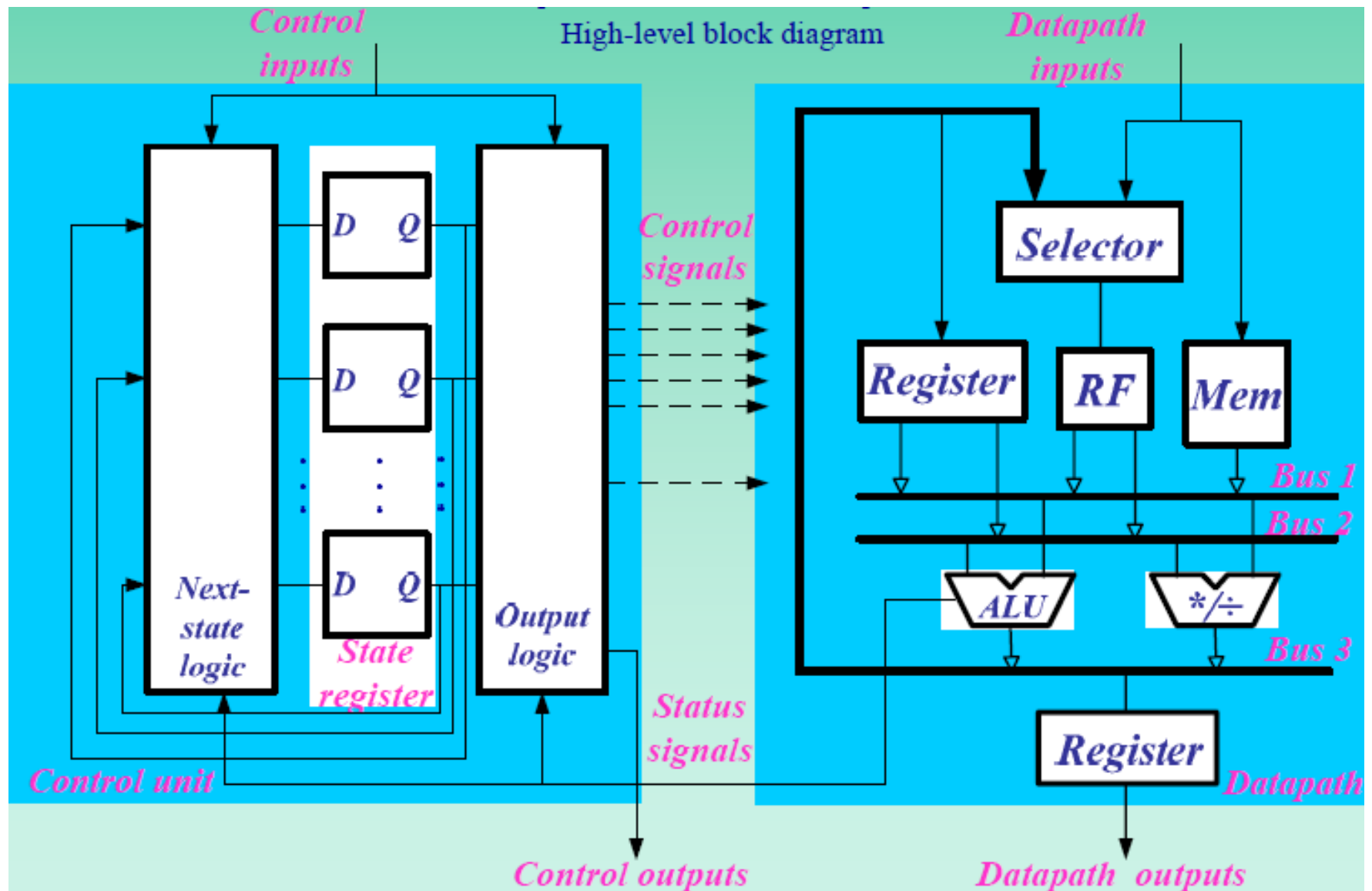
$$Sh_0 = 0$$

$$WE = Q_2 Q'_1 + Q'_2 Q_0 + Q_1 Q_0$$

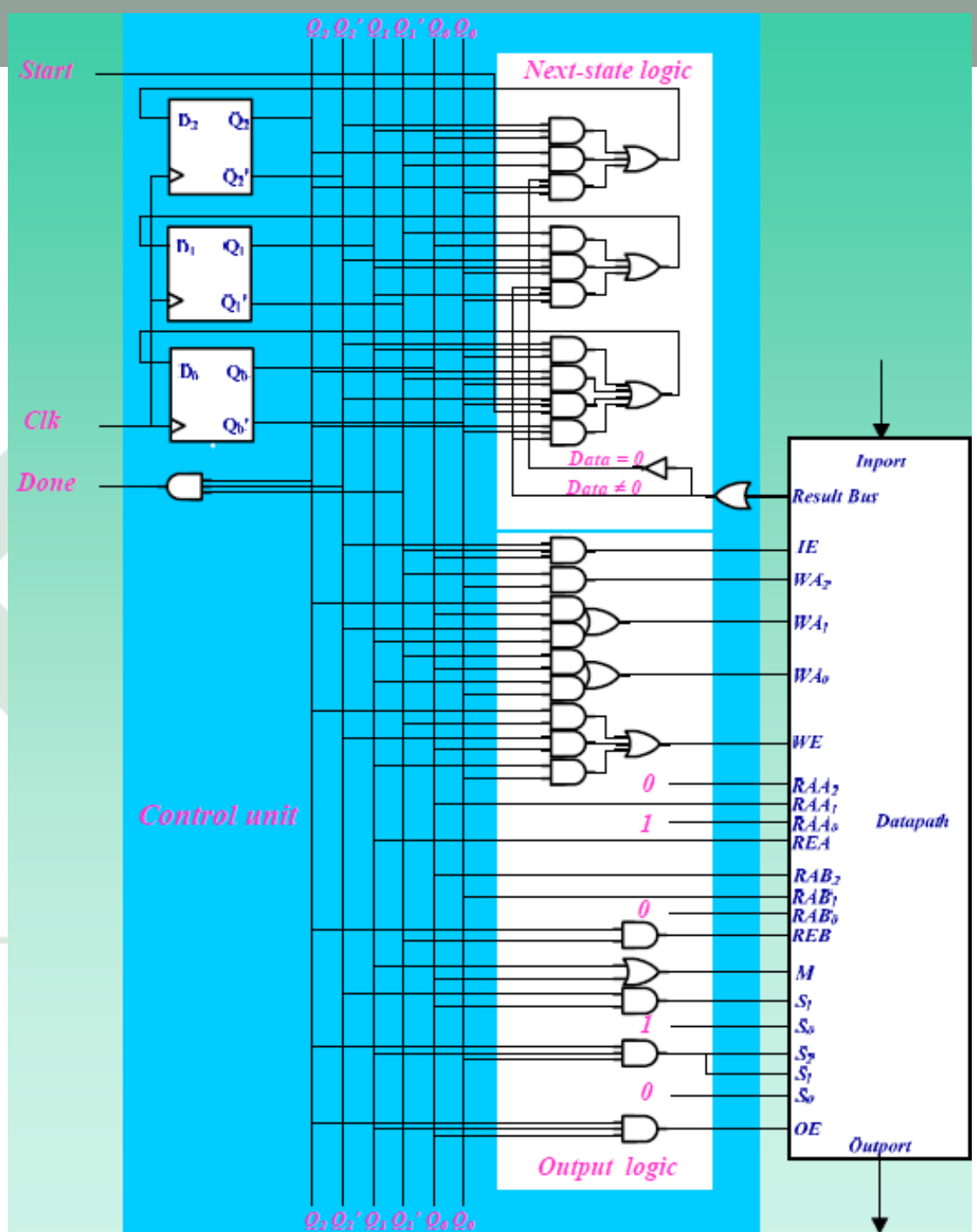
$$OE = Q_2 Q_1 Q_0$$

Sách sai

Design model



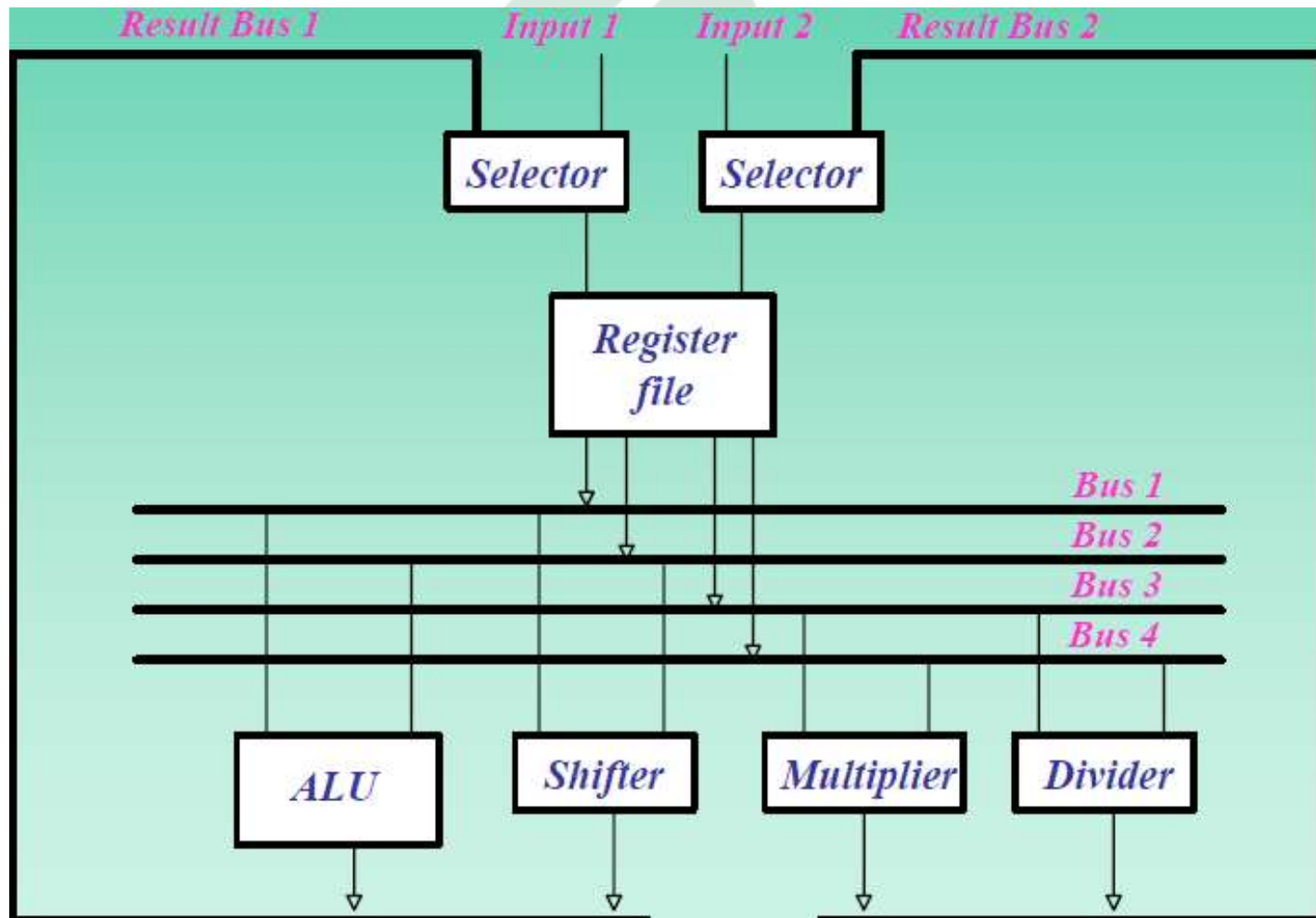
One's-counter schematic



Mạch bị sai tại các dòng
màu xanh trong slide 10

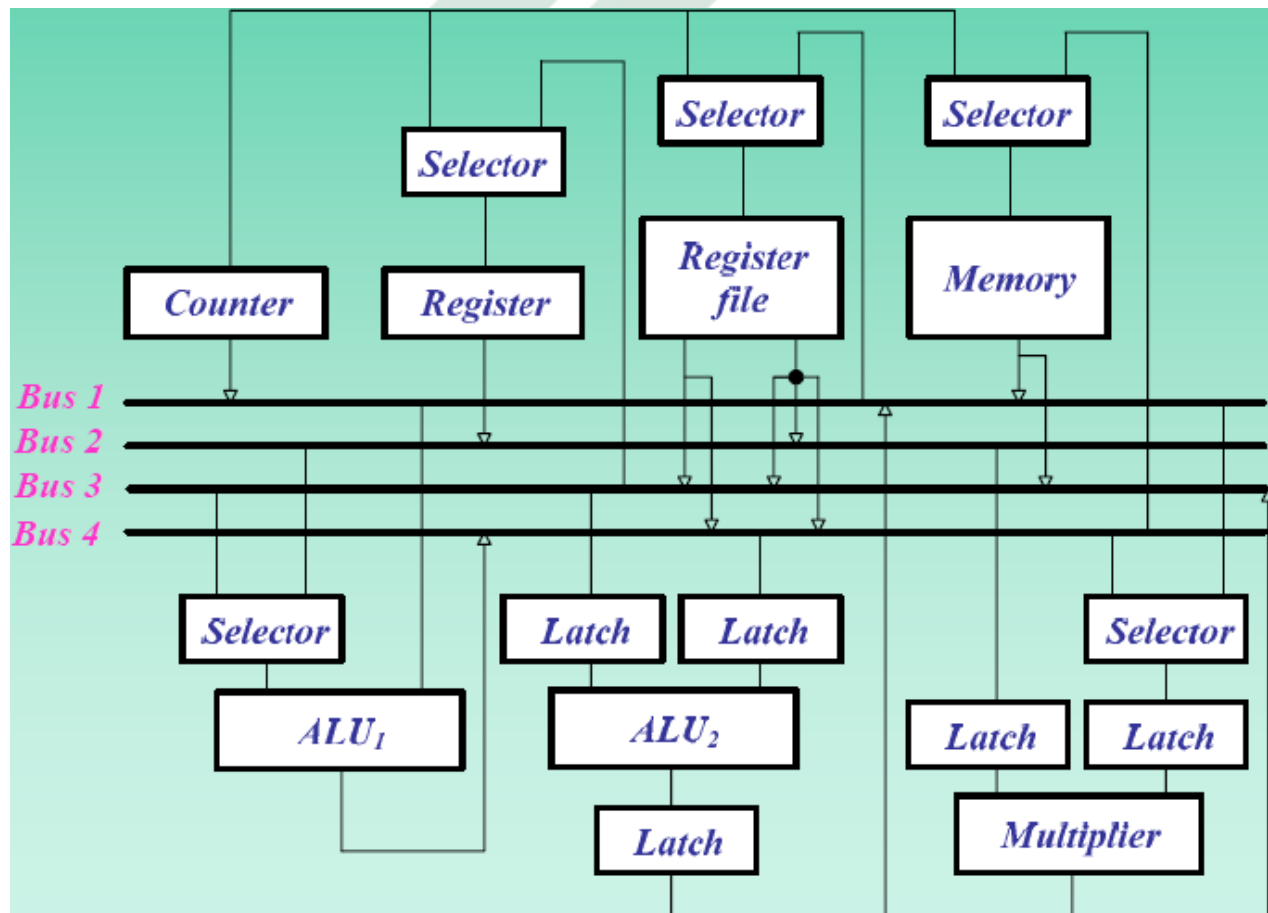
General Datapath - Parallel datapath

- Datapath allows 2 operations in each clock cycle.
- Choice of operations performed simultaneously is limited.

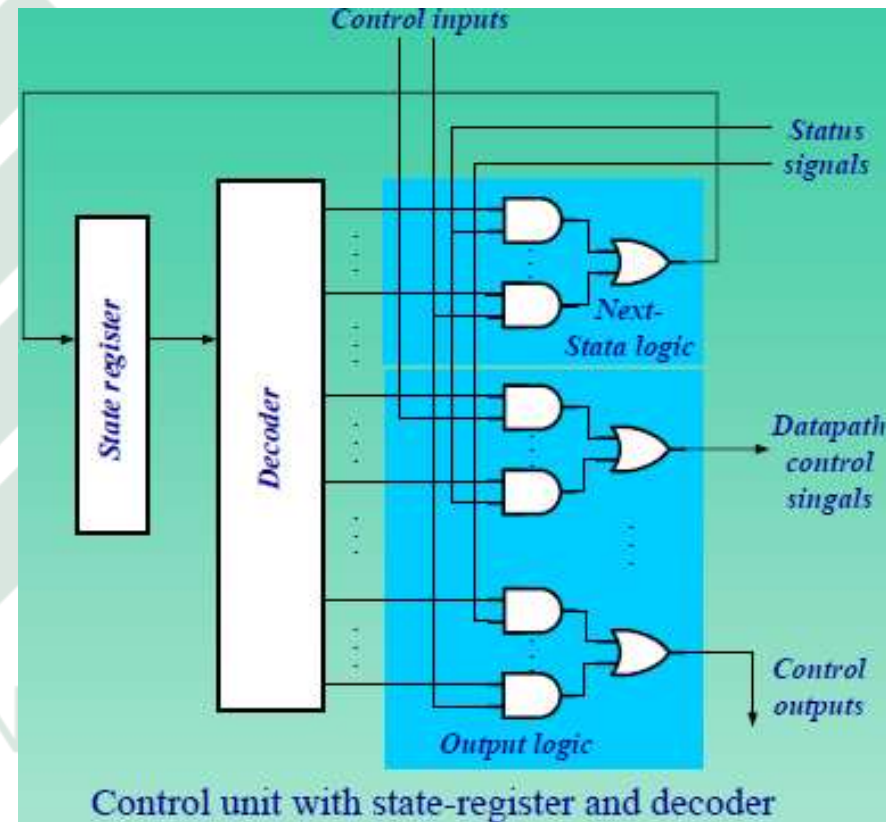
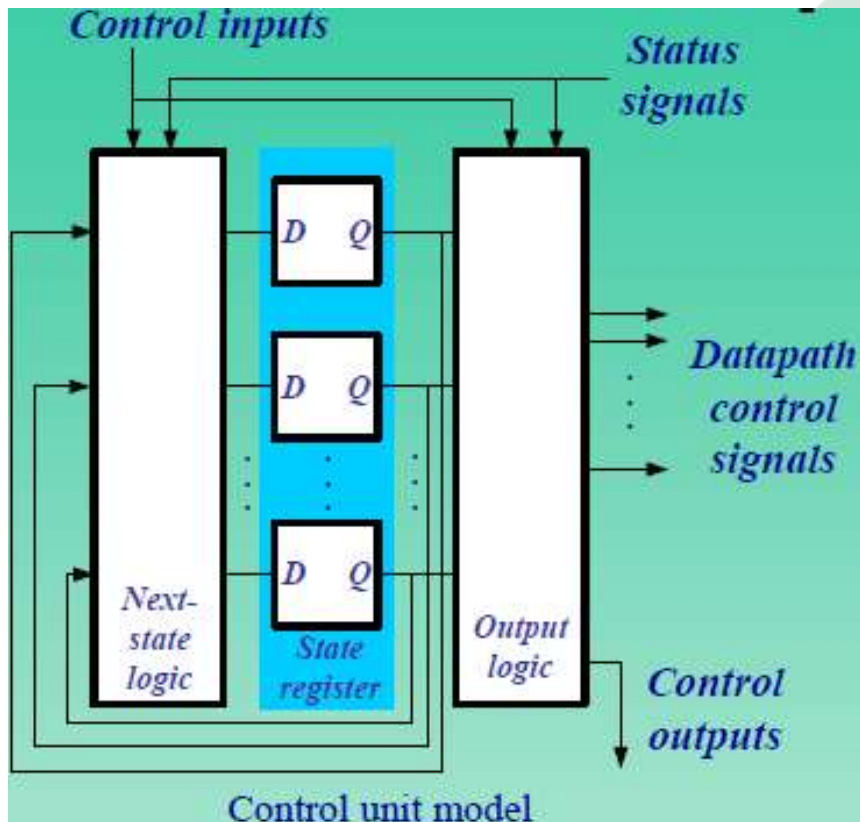


An example of a custom datapath

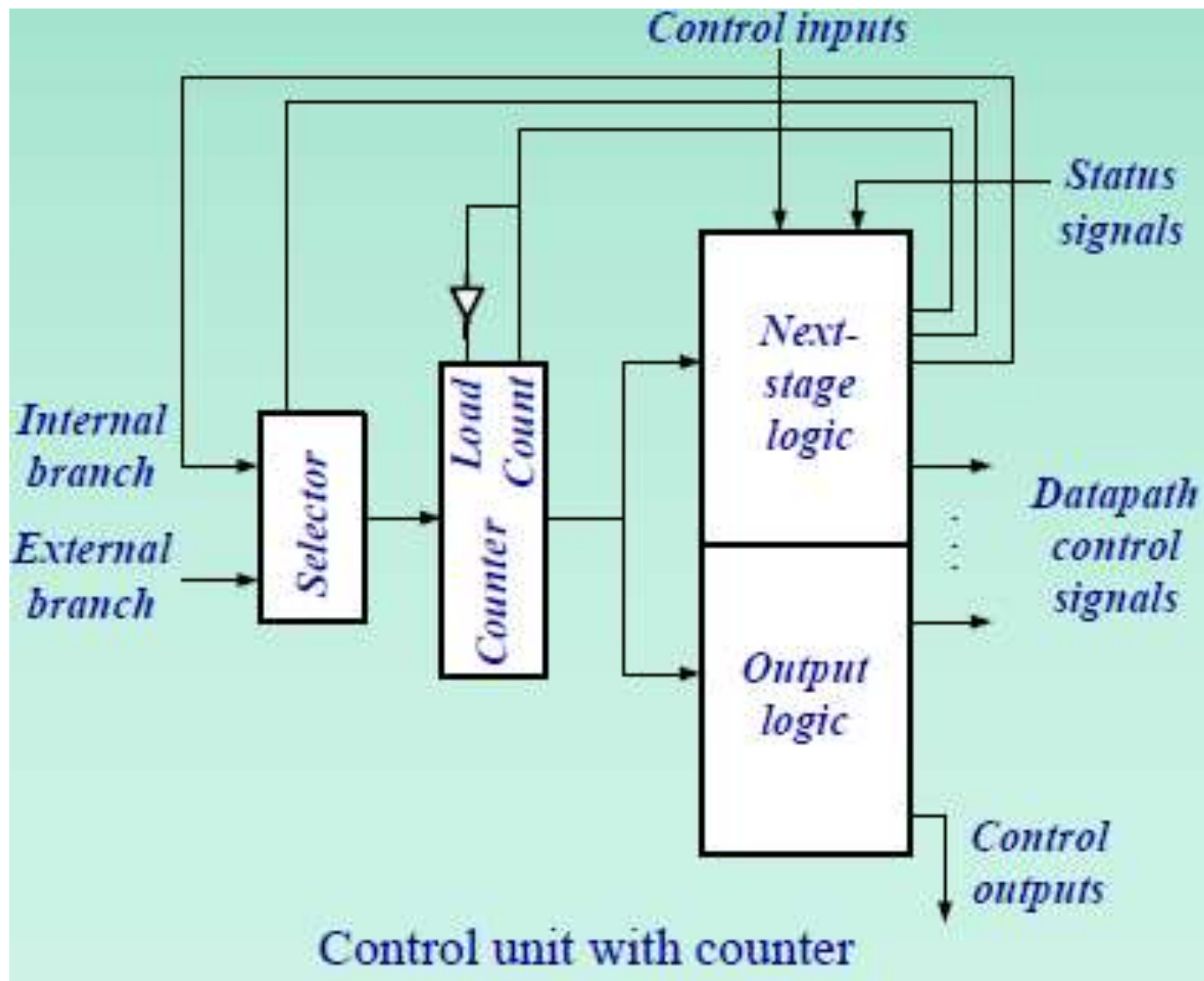
- Uses latches as temporary storage on the input and output of functional units.
- Latches allow shorter clock cycles and more concurrency.



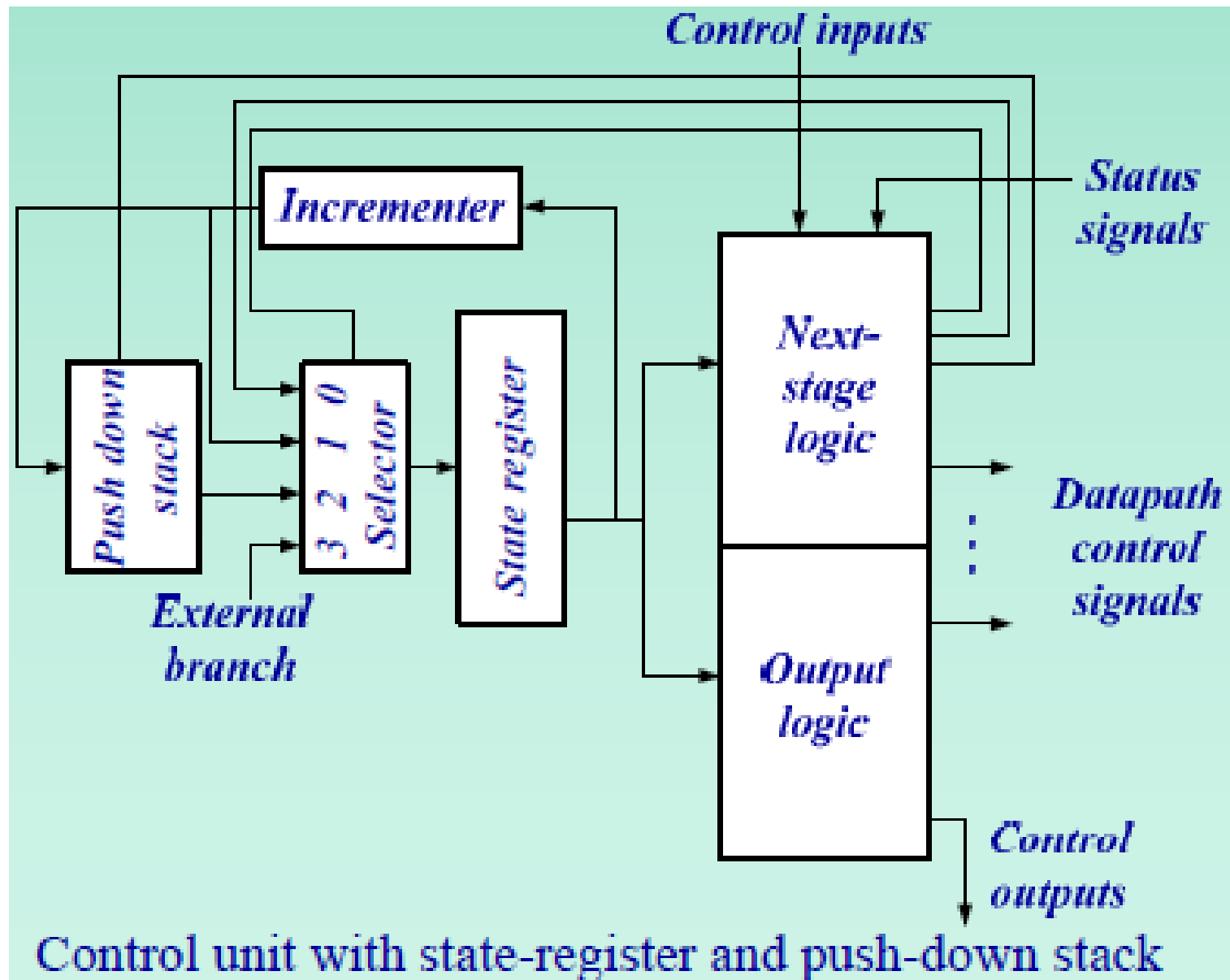
Control-unit implementation styles



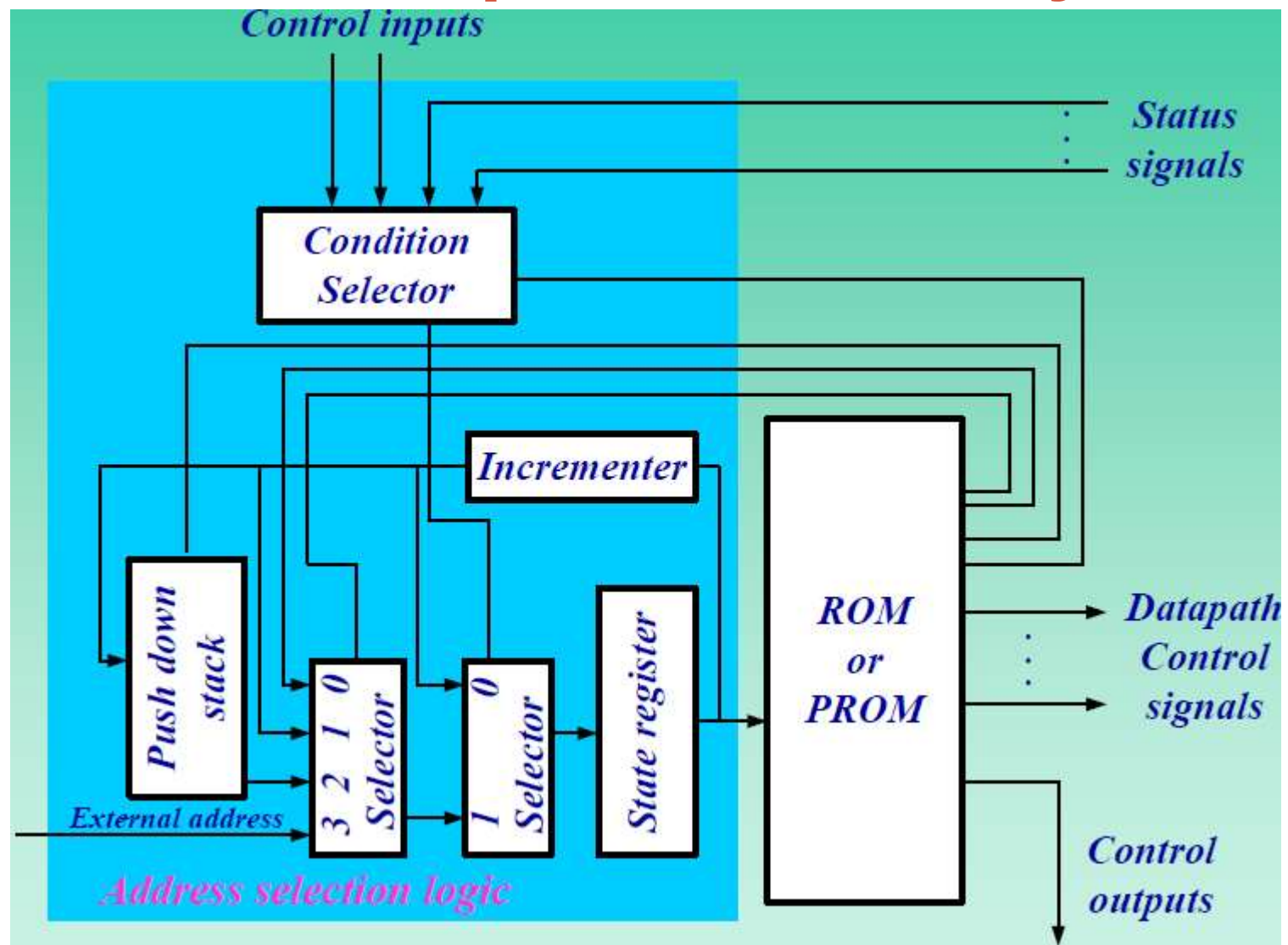
Control-unit implementation styles



Control-unit implementation styles



Control-unit implementation styles



Control unit with state-register, ROM and push-down stack
(microprogrammed control unit)

Chapter Summary

- We introduced sequential components
 - Simple (registers, counters)
 - Storage (register-files, memories)
 - Complex (stacks, queues, datapaths and control units)
- Datapath and control units are used to implement standard processors and application specific integrated circuits (One's count example)

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Bài tập

➤ Thiết kế datapath thực hiện công việc sau:

$$S = 1 + 2 + 3 + \dots + n$$

Với n là dữ liệu do người dùng nhập vào



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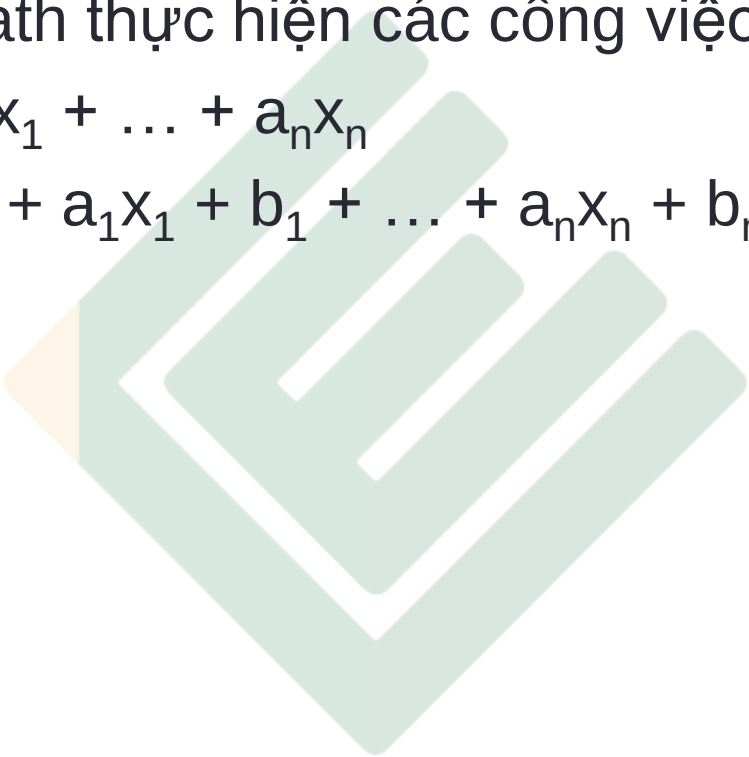
Bài tập

➤ Thiết kế datapath thực hiện các công việc sau:

a. $S = a_0x_0 + a_1x_1 + \dots + a_nx_n$

b. $S = a_0x_0 + b_0 + a_1x_1 + b_1 + \dots + a_nx_n + b_n$

Với các tham số



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Bài tập

➤ Thiết kế datapath thực hiện công việc sau:

- Cho 2 số A, B có độ dài 8 bit: $A = a_7a_6a_5a_4a_3a_2a_1a_0$; $B = b_7b_6b_5b_4b_3b_2b_1b_0$
- Kết quả trả về C như sau: $C = b_7b_6a_3a_2a_1a_0b_1b_0$

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